REAL TIME SAFETY MONITORING SYSTEM FOR PUBLIC TRANSPORT

Project Group: 25-26J-499

Project Proposal Report

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Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

REAL-TIME BUS TRACKING SYSTEM WITH OCCUPANCY AND TIME PREDICTION

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
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The supervisor/s should certify the proposal report with the following declaration.
The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: Date 25/08/2025

ABSTRACT

Public transportation in Sri Lanka, primarily operated by the Sri Lanka Transport Board (SLTB) and private bus companies plays a vital role in the mobility of millions of commuters. However, passengers face recurring issues such as uncertain bus arrival times, overcrowding, and lack of predictive journey information. While fleet management solutions exist locally (e.g., Kloudip, Kommnet), they remain operator-oriented and fail to provide commuter-facing predictive insights. Globally, predictive transport systems have demonstrated the effectiveness of statistical models such as ARIMA, regression and hybrid methods, as well as machine learning and deep learning approaches including Random Forests, LSTMs, GRUs, and CNN-based architectures. These models have been successfully applied for short-term traffic forecasting, bus arrival time prediction, and congestion modeling. Similarly, research on occupancy prediction has leveraged APC data, smart card datasets, and passenger flow modeling to forecast bus crowding with high accuracy. This research proposes a predictive real-time bus monitoring and passenger information system designed for SLTB and private operators. Using operational datasets (GPS logs, timetables, and ticketing records) from SLTB, machine learning models will be developed to forecast both arrival times and occupancy levels. The system will integrate real-time GPS tracking with predictive analytics, delivering insights to passengers via a mobile application. By shifting from simple real-time tracking to predictive, data-driven analytics, this research aims to enhance commuter trust, reduce waiting time, and improve comfort while supporting operators and transport authorities with fleet optimization and policy planning.

Keywords: Public Transport, Real-Time Tracking, Arrival Time Prediction, Occupancy Forecasting, SLTB, Machine Learning, Smart Transport Systems

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LIST OF ABBREVIATIONS

Abbreviation	Description
GPS	Global Positioning System
SLTB	Sri Lanka Transport Board
ARIMA	AutoRegressive Integrated Moving Average
LSTM	Long Short-Term Memory (neural network)
GRU	Gated Recurrent Unit (neural network)
CNN	Convolutional Neural Network
ML	Machine Learning
IoT	Internet of Things
APC	Automatic Passenger Counting
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
API	Application Programming Interface
SaaS	Software as a Service
AVL	Automatic Vehicle Location
GIS	Geographic Information System

1. INTRODUCTION

The efficiency and reliability of Sri Lanka's public transport network are critical for both socio-economic development and the daily mobility of millions of commuters. Buses, operated by the state-owned Sri Lanka Transport Board (SLTB) [1] and private bus operators, form the backbone of land-based public transportation in the country. Despite their importance, recurring issues such as uncertain bus arrival times, irregular schedules, overcrowding, and lack of reliable passenger information continue to undermine service quality [2][5][25]. These issues contribute to reduced commuter satisfaction, driving passengers toward private transport, which in turn increases congestion and environmental stress [16][24].

Current systems in Sri Lanka largely focus on limited real-time tracking or operator monitoring [3][4]. The absence of predictive analytics leaves passengers without actionable information such as accurate bus arrival forecasts or occupancy levels. Consequently, passengers face long waiting times at bus stops, overcrowded conditions, and overall inconvenience, reducing trust in public transport.

To address these challenges, this research proposes a **data-driven predictive passenger information system** that combines real-time bus tracking with machine learning–based arrival time and occupancy prediction. Leveraging operational datasets from SLTB (GPS logs, schedules, and ticketing records) [1][25], models such as ARIMA, Random Forest, and LSTM [6][7][10][21][23] will be applied to forecast arrival times and passenger load. These predictions will be integrated into a commuter-facing mobile application, providing real-time and predictive insights to passengers. By shifting from reactive tracking to predictive analytics, the proposed system aims to improve reliability, convenience, and commuter confidence while supporting SLTB and private operators in optimizing fleet management [14][15][19].

1.1 Background and Literature Survey

1.1.1 Background on Sri Lankan Bus Transportation and Policy

Sri Lanka's public bus system connects rural and urban areas, ensuring accessibility for millions daily [1]. The dual-operator structure of SLTB and private bus companies offers nationwide coverage, yet service quality remains inconsistent due to irregular schedules, overcrowding, and lack of commuter-oriented systems [2][25]. Although SLTB has introduced digital ticketing and route scheduling [1], these initiatives are underutilized for predictive analytics.

Passenger dissatisfaction in Sri Lanka arises mainly from (i) unpredictable bus arrival times and long waiting periods, and (ii) overcrowding that reduces comfort and safety [5][9]. While operator-focused platforms exist, such as Kloudip [3] and Kommnet [4], their services are centered on fleet tracking, fuel monitoring, and vehicle performance, rather than providing predictive information to passengers. The Sri Lankan government has highlighted modernization through policies that include smart ticketing, IoT-enabled monitoring, and digital route management [2]. However, these initiatives are still limited in providing actionable predictions for passengers.

Thus, despite infrastructure digitization, the **absence of predictive public transport systems** represents a missed opportunity to use available SLTB datasets for commuter-facing analytics.

1.1.2 Review of Existing Solutions and Academic Literature

1.1.2.1 Local Systems and Limitations

Several private companies in Sri Lanka offer GPS-based fleet tracking solutions for logistics and bus operators [3][4]. While effective for monitoring vehicles, fuel efficiency, and scheduling, these solutions are not designed for passengers. Real-time arrival predictions or occupancy forecasts are not provided, leaving commuters uninformed and relying on guesswork.

1.1.2.2 Global Academic Research on Predictive Passenger Information Systems

Globally, extensive research has demonstrated that predictive models significantly improve public transport services. Traditional time-series and regression approaches [6][12][20] have been used for arrival forecasting, while modern deep learning models such as LSTM, GRU, and hybrid CNN-RNN architectures [7][10][11][18][21][23] provide higher accuracy by capturing non-linear and temporal traffic patterns. Research by Chen et al. [6] and Ma et al. [7] shows that deep learning can effectively predict bus arrival under congestion and dynamic conditions.

Occupancy forecasting has also been widely studied. Approaches using smart card and APC data [8][9][17][22] demonstrate accurate predictions of passenger loads. For example, Rahbar et al. [9] developed real-time crowding prediction using ML in Stockholm, while Moreira-Matias et al. [19] applied streaming passenger demand models for large-scale transport networks. Such systems improve passenger satisfaction, operator scheduling, and safety.

Although predictive transport research is well established internationally, most solutions rely on advanced infrastructure and smart city environments [11][18]. Sri Lanka requires a **cost-effective approach leveraging SLTB's existing datasets** without heavy hardware investments.

1.2 Research Gap

There are no existing Sri Lankan telematics platforms that provide basic real-time tracking for operators and predictive analytics to passengers. On the other hand, international research demonstrates sophisticated predictive systems, but they are rarely applied in resource-constrained contexts like Sri Lanka.

The specific gaps are:

Area	Existing Local Systems (Sri Lanka)	Global Academic Research	Identified Gap
Real-Time Tracking	Limited to fleet management tools (e.g., Kloudip, Kommnet); not commuter-oriented.	Widely available in smart city projects.	No commuter-facing applications in Sri Lanka providing live arrival data.
Arrival Time Prediction	Not implemented; current systems only show live GPS location.	Advanced models (ARIMA, LSTM, GRU, hybrid models) used for accurate forecasting.	Lack of predictive analytics in Sri Lanka's bus systems.
Occupancy Prediction	Completely absent in Sri Lanka's systems.	Occupancy predicted using smart card/ticketing data and ML models.	No mechanism for predicting bus crowding or passenger load locally.
Use of SLTB Data	SLTB digitizes ticketing/schedules but rarely used for passenger services.	Historical datasets widely used in research to improve accuracy.	SLTB's data not leveraged for commuter-facing predictive insights.

Passenger-Facing Applications	Mostly reactive; limited tracking features, no prediction.	Fully integrated apps provide real-time + predictive insights.	No unified predictive system available for passengers in Sri Lanka.

Table 1.2: Research gap comparison table

This highlights the need for a scalable, commuter-focused predictive system tailored to Sri Lanka's public transport environment.

1.3 Research Problem

The central research problem is the **lack of an integrated predictive transport information system in Sri Lanka**. Current solutions either serve operators through fleet monitoring or provide only GPS-based tracking to passengers. Without predictive insights, commuters remain uninformed about when buses will arrive or how crowded they will be.

International research demonstrates that predictive systems enhance reliability, reduce waiting times, and increase trust in public transport. However, Sri Lanka has yet to adopt such solutions in practice.

This research aims to fill this gap by developing a **machine learning–driven framework** that integrates SLTB's operational datasets with predictive models for **bus arrival time and occupancy forecasting**. The proposed commuter-facing mobile app and operator dashboard will enhance passenger convenience, fleet reliability, and data-driven transport planning in Sri Lanka.

2. OBJECTIVES

2.1. Research Questions

To address the identified research problem, this project will seek to answer the following key questions:

- How can SLTB's GPS and operational datasets be preprocessed and modeled to accurately predict bus arrival times in real-world conditions?
- What machine learning or statistical techniques (e.g., regression, ARIMA, LSTM, or hybrid models) are most effective for forecasting bus occupancy levels?
- How can real-time tracking and predictive insights be integrated into a **passenger-facing mobile application** in a scalable and user-friendly way?
- What communication and system architecture will ensure **low-latency updates** for both live tracking and predictive information?

2.2. Main Objective

The main objective of this research is to **design, develop, and validate a predictive real-time bus tracking and passenger information system** that uses SLTB data to provide commuters with accurate **arrival time predictions and occupancy forecasts**, thereby improving efficiency, reliability, and user satisfaction in Sri Lanka's public transport sector.

2.3. Specific Objectives

- **Collect and Preprocess SLTB Data**
  Gather and clean historical GPS, trip, and ticketing datasets from SLTB for model training and evaluation.
- **Develop Bus Arrival Time Prediction Models**
  Build machine learning models (e.g., regression, ARIMA, LSTM) capable of forecasting bus arrival times with high accuracy.
- **Build Occupancy Prediction Models**
  Use passenger ticketing data and historical ridership patterns to predict occupancy levels, helping commuters plan for crowded or empty buses.
- **Develop a Passenger-Facing Mobile Application**
  Create a mobile app that integrates real-time tracking with predicted arrival times and occupancy, presented in an intuitive and user-friendly interface.
- **Validate and Evaluate System Performance**
  Conduct accuracy testing of predictive models, usability testing of the passenger application, and end-to-end evaluation of the system's reliability under real operating conditions.

3. METHODOLOGY

The methodology for this research follows three main phases: **data acquisition and preprocessing, model development and system design, and validation and testing**. This structured approach ensures the final solution is technically robust, accurate, and practical for large-scale deployment in Sri Lanka's public transport system.

3.1 Data Collection and Preprocessing

The primary dataset will be obtained from the Sri Lanka Transport Board (SLTB). Data sources include:

- **GPS logs** – real-time bus movement data including route, distance, and time.
- **Trip sheets and timetables** – historical records of scheduled versus actual departure/arrival times.
- **Ticketing data** – passenger boarding counts and occupancy trends.

Preprocessing steps:

- Cleaning and handling missing values in GPS and trip data.
- Standardizing timestamps to align real-time and historical datasets.
- Feature engineering, e.g., calculating travel intervals, peak/off-peak tags, and weather/day-type attributes.
- Splitting datasets into training, validation, and testing subsets for predictive model development.

3.2 Prediction Model Development

The predictive framework will consist of two modules:

a) Arrival Time Prediction

- Supervised machine learning and statistical models such as Multiple Linear Regression, ARIMA, Random Forests, and LSTMs.

- Features include GPS speed, route distance, congestion patterns, and historical delays.
- Evaluation metrics: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

b) Occupancy Prediction

- Models trained using ticketing data, boarding/alighting patterns, and temporal factors (time of day, weekday/weekend, holidays).
- Approaches: Regression models, Time-Series forecasting, and Neural Networks.
- Evaluation metrics: Accuracy, Precision, Recall, and Confusion Matrix for predicted vs actual occupancy levels.

Both models will be tested independently and then integrated to provide passengers with **combined real-time and predictive information**.

3.3 System Architecture

The system will adopt a three-tier architecture:

1. **Data Layer**
    - SLTB data stored and processed in a cloud-hosted environment (e.g., Firebase, AWS, or Google Cloud).
    - APIs for real-time GPS updates and historical dataset access.
2. **Processing Layer**
    - Machine learning models trained on historical datasets and deployed on the backend.
    - Continuous updates as real-time GPS and new ticketing data are fed into the models.
3. **Application Layer**
    - **Passenger Mobile App**: Provides live bus tracking, estimated arrival times, and predicted occupancy levels.
    - **Operator Dashboard**: Allows SLTB staff to monitor predictions and evaluate route efficiency.

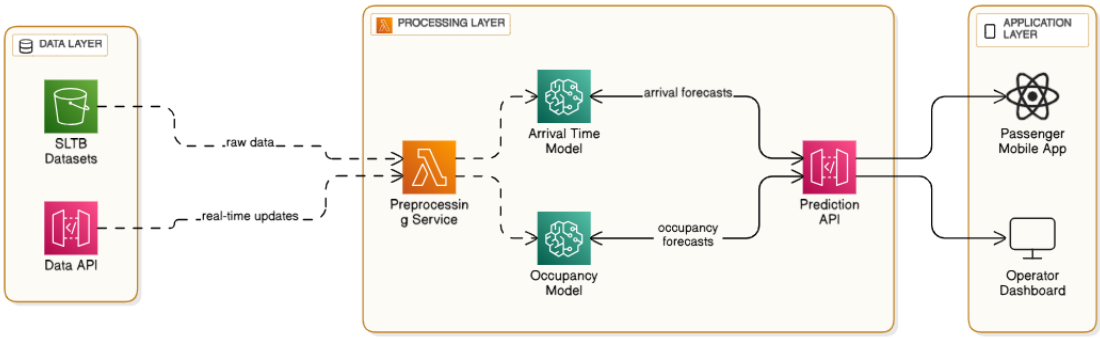

Figure 3.2: System diagram for the component

3.4 Validation and Testing

The system will be evaluated through multiple testing stages:

- **Unit Testing**: Validate individual model components (arrival time prediction, occupancy forecasting).
- **Functional Testing**: Ensure predictions are correctly integrated with the mobile application.
- **Accuracy Testing**: Compare predicted results with actual arrival times and passenger counts.
- **User Testing**: Deploy pilot versions to a small commuter group for usability and satisfaction feedback.
- **End-to-End Testing**: Simulate full-scale operation with multiple routes to confirm real-time integration and system scalability.

3.5 Tools and Technologies

- **Data Processing & Modeling**: Python, Pandas, Scikit-Learn, TensorFlow/Keras, StatsModels (for ARIMA).
- **Backend**: Node.js / Firebase (for APIs and data synchronization).
- **Mobile Application**: React Native (cross-platform support).
- **Database**: Cloud Firestore or MySQL for structured trip and passenger data.

4. DESCRIPTION OF PERSONNEL AND FACILITIES

4.1 Personnel

This project will be undertaken by a single student researcher:

Name: Samarakoon S.M.J.B

Role: Researcher & Developer

Responsibilities:

- Collection and preprocessing of SLTB datasets (GPS, timetables, ticketing).
- Development and training of machine learning models for arrival time and occupancy prediction.
- Design and implementation of the mobile application with real-time tracking and predictive features.
- Backend development for model integration, data handling, and API services.
- System testing, validation, and performance evaluation.
- Documentation and reporting.

The student will be supervised by an academic supervisor who will:

- Provide technical and academic guidance on data analytics and system design.
- Review progress and ensure adherence to research objectives.
- Assist with validation approaches and model evaluation techniques.

4.2 Facilities

The project will utilize the following facilities and resources:

University Facilities:

- Access to computing resources and licensed software tools through SLIIT.
- Access to cloud platforms for hosting datasets and running predictive models.

- Guidance from faculty on research methodology and machine learning.

Personal Resources:

- Personal laptop for model development, coding, and testing.
- Internet access for research, collaboration, and remote data handling.
- Android smartphone for testing the mobile application.

Software & Tools:

- Python (Pandas, Scikit-Learn, TensorFlow/Keras, StatsModels) for data analysis and predictive modeling.
- Node.js / Firebase for backend services and data synchronization.
- React Native for mobile application development.
- Cloud database solutions (Firebase Firestore) for structured data storage and retrieval.

5. BUDGET AND BUDGET JUSTIFICATION

Item	Quantity	Unit Cost (LKR)	Total Cost (LKR)	Justification
Cloud Hosting (Firebase)	1 year	6,000	6,000	Required for storing datasets, running backend services, and deploying APIs for real-time updates.
Data Processing & ML Tools	1 set	5,000	5,000	Includes Python libraries, Jupyter environment setup, and cloud-based ML processing credits.
Mobile Application Development Tools & Licenses	1 set	5,000	5,000	Required for cross-platform app development using React Native and testing frameworks.
Database & Storage Services	1 year	4,000	4,000	Cloud-based database for storing SLTB data, prediction results, and user information.
Testing Devices (Android Smartphone)	1	20,000	20,000	For developing and testing the passenger-facing mobile application.

Miscellaneous (Internet, printing, documentation, small-scale deployment costs)	Lump sum	—	5,000	Covers minor expenses related to research, reporting, and app testing.
Total Estimated Cost	—	—	45,000	Overall estimated cost for the project

Table 5 : Budget justification table

Budget Notes:

- Most development tools are open-source (Python, React Native), reducing software licensing costs.
- Primary expenses are related to **cloud hosting, database services, and mobile app development/testing**.
- The budget reflects a **student research project** with minimal upfront investment, while ensuring scalability for future deployment across SLTB and private operators.

6. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed predictive real-time bus tracking and passenger information system has significant commercial potential within Sri Lanka's public transport sector. The increasing demand for smarter, more reliable, and data-driven solutions provides an opportunity to deploy the system across both **state-run SLTB buses** and **private operators**.

Target Customers

- **Sri Lanka Transport Board (SLTB):** As the largest bus operator in the country, SLTB can adopt this system to improve service reliability, and strengthen commuter confidence.
- **Private Bus Operators:** With thousands of private buses operating islandwide, the system can be offered as a scalable solution to improve competitiveness and provide value-added services to passengers.

Value Proposition

- **For Passengers:** Access to real-time bus locations, predicted arrival times, and expected occupancy, enabling better travel decisions and reducing waiting times.
- **For Operators (SLTB & Private):** reduced passenger complaints, data-driven insights for scheduling, and optimized resource allocation.
- **For Authorities:** Availability of predictive data to support traffic management, route planning, and long-term transport policy improvements.

Business Model

The system can be commercialized using a **Software-as-a-Service (SaaS)** model:

- **Subscription-based Pricing:** Operators pay a monthly or annual fee per bus to access the platform.
- **Features included in the subscription:**
    - Passenger-facing mobile application with live tracking and predictions.
    - Backend data processing and prediction services.

    - Access to the operator dashboard for fleet monitoring and performance analytics.
    - Continuous updates and technical support.

Scalability & Market Expansion

- **National Scale:** Initially deployed for SLTB fleets, followed by expansion to private operators in urban and rural areas.
- **Regional Expansion:** The system can be adapted for other South Asian countries facing similar issues of overcrowding, irregular bus services, and lack of predictive passenger information.

Commercial Benefits

- Enhances **public trust** in bus services by reducing uncertainty.
- Provides **competitive differentiation** for private operators adopting the system.
- Creates **new revenue streams** through predictive data services, mobile app features, and potential partnerships with ticketing/payment platforms.

7. PROJECT REQUIREMENTS

The success of the proposed system depends on defining clear functional and non-functional requirements. These requirements ensure that the solution meets its goals of providing accurate **real-time tracking, arrival time prediction, and occupancy forecasting** to passengers and operators.

7.1 Functional Requirements

1. **Real-Time Bus Tracking**
    - Provide continuous GPS updates to track the live location of buses.
    - Display route progress and estimated arrival times for upcoming stops.
2. **Arrival Time Prediction**
    - Use historical and real-time SLTB data to forecast bus arrival times.
    - Continuously update predictions as traffic and route conditions change.
3. **Occupancy Prediction**
    - Forecast passenger load based on historical ticketing data, time of day, and day type (weekday/weekend/holiday).
    - Display expected occupancy levels (e.g., Low, Medium, High) to passengers.
4. **Passenger-Facing Mobile Application**
    - Provide a user-friendly interface showing live bus location, arrival predictions, and occupancy forecasts.
    - Support push notifications for schedule delays or major changes.
5. **Operator Dashboard**
    - Provide SLTB and private operators with analytics on predicted vs actual times and occupancy.
    - Support route-level performance monitoring.
6. **Data Logging and Storage**
    - Store GPS, prediction outputs, and passenger usage data securely for continuous model improvement and compliance reporting.

7.2 Non-Functional Requirements

- **Performance:** Ensure predictions update in real-time with minimal latency (<5 seconds).
- **Accuracy:** Maintain acceptable error margins for predictions (e.g., ≤3 minutes for arrival time forecasts, ≥80% accuracy for occupancy prediction).
- **Reliability:** System should have 99.9% uptime with failover support in cloud hosting.
- **Security:** Encrypt passenger and operational data; enforce secure authentication for operators.
- **Scalability:** The system must support thousands of buses simultaneously, across both SLTB and private operators.
- **Usability:** Interfaces should be intuitive, requiring minimal training for passengers and operators.

7.3 User Requirements

High-level requirements from the **end-users' perspective** (passengers, operators).

- **Passengers**
    - Passengers should be able to view the **real-time location** of buses on a mobile application.
    - Passengers should receive **predicted bus arrival times** with acceptable accuracy.
    - Passengers should be informed of **bus occupancy levels** (Low, Medium, High) before boarding.
    - Passengers should receive **alerts/notifications** for delays, breakdowns, or schedule changes.
    - Passengers should be able to use a **simple, user-friendly mobile interface**.
- **Operators (SLTB & Private Bus Companies)**

- Operators should be able to **monitor their fleet** in real-time.
- Operators should be able to view **analytics reports** (predicted vs actual times, occupancy trends).
- Operators should have **secure access** to system dashboards and data.

7.4 Expected Test Cases

1. **Real-Time Location Tracking**
   - Verify that live bus locations update correctly in the passenger app.
2. **Arrival Time Prediction Accuracy**
   - Compare predicted vs actual bus arrival times across multiple test routes.
3. **Occupancy Prediction Accuracy**
   - Validate predicted passenger load against actual boarding counts.
4. **Mobile App Usability**
   - Ensure predictions and tracking are displayed clearly and intuitively for passengers.
5. **Operator Dashboard Validation**
   - Test data analytics features such as predicted vs actual route performance.
6. **End-to-End System Stability**
   - Simulate large-scale operation across multiple routes to validate scalability and reliability.

8. WORK BREAKDOWN STRUCTURE

8.1 Project Timeline and Resource Allocation

Phase	Task ID	Task Description
Phase 1: Data Acquisition and Preprocessing	1.1	Collect GPS, trip sheets, and ticketing data from SLTB
	1.2	Clean and preprocess raw data (handle missing values, outliers, and time alignment)
	1.3	Perform feature engineering (peak/off-peak tagging, route congestion factors, holiday/weekend classification)
	1.4	Split datasets into training, validation, and testing subsets
Phase 2: Model Development	2.1	Develop bus arrival time prediction models using Regression, ARIMA, Random Forest, and LSTM
	2.2	Train and evaluate models for occupancy forecasting using ticketing and ridership data
	2.3	Optimize models for accuracy, efficiency, and real-time applicability
	2.4	Integrate prediction modules into the backend service

Phase 3: System Design & Development	3.1	Design overall system architecture (data layer, processing layer, application layer)
	3.2	Implement backend server and APIs for real-time data handling
	3.3	Develop mobile application (React Native) featuring real-time tracking and predictive insights
	3.4	Develop operator dashboard for SLTB and private fleet managers
Phase 4: Testing and Validation	4.1	Conduct unit testing of prediction models
	4.2	Perform functional testing of the mobile app (live tracking, arrival time, and occupancy forecast)
	4.3	Test accuracy of arrival time and occupancy predictions against actual SLTB trips
	4.4	Conduct end-to-end system testing using pilot routes
Phase 5: Documentation and Deployment	5.1	Prepare comprehensive project documentation (proposal, progress reports, final thesis)
	5.2	Deploy prototype system for trial with selected SLTB routes

	5.3	Collect feedback from users and operators to inform improvements
	5.4	Prepare commercialization strategy and final project presentation

Table 8.1: Project Timeline and Resource Allocation

8.2 Grant chart

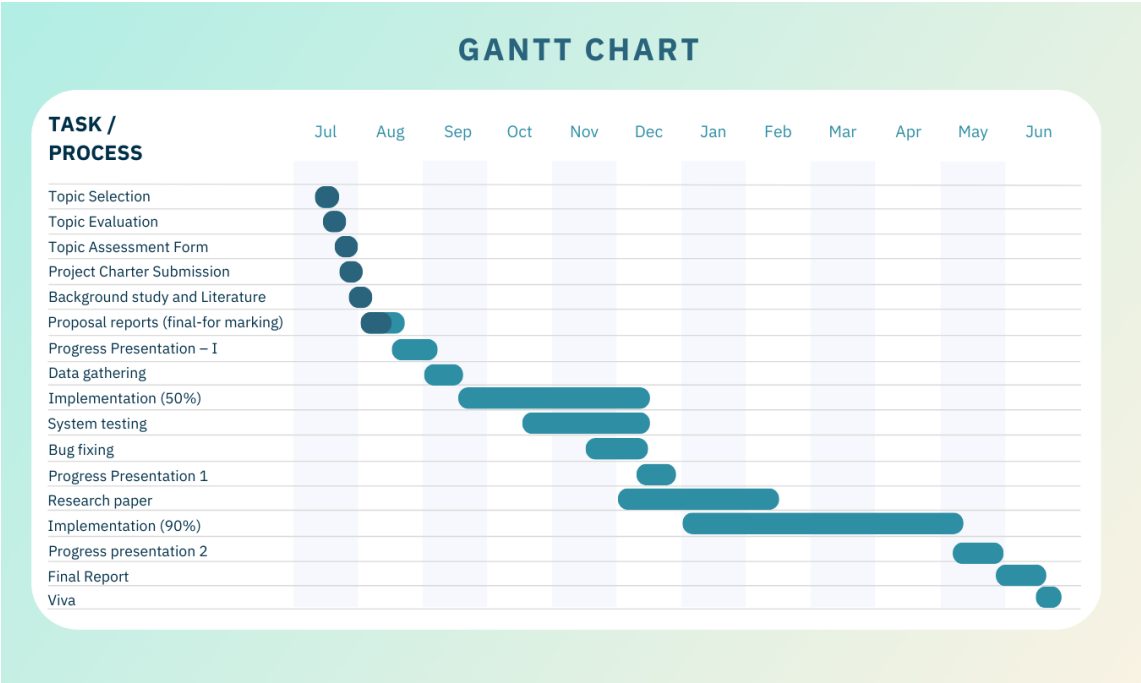

Figure 8.1: Grant Chart

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APPENDICES

The General Manager

Sri Lanka Transport Board (SLTB)

Request for Dataset Access for Research Project on "Safety Monitoring System for Public Transport in Sri Lanka"

Dear Sir

I am writing on behalf of the Sri Lanka Institute of Information Technology (SLIIT) to kindly request access to relevant datasets from your esteemed organization to support an academic research project.

Our research project, titled "Safety Monitoring System for Public Transport in Sri Lanka," aims to investigate, design, and develop a digital solution to enhance safety standards in public transportation. To ensure the accuracy and effectiveness of our findings, we require access to vehicle operation data, accident/incident reports, maintenance records, GPS/telemetry data, and any other safety-related datasets available within your organization.

The requested data will be used solely for academic and research purposes and will remain confidential. All findings and outcomes will be shared with your organization upon completion of the project.

We would be grateful for your kind cooperation in granting us the necessary datasets to facilitate this study.

Thank you very much for your valuable time and consideration. We look forward to your positive response.

Yours faithfully,

Research Student(s):

Name: .....

Reg. No: .....

Supervisor:

.....

Thisara Shyamalee

Lecture at SLIIT

Co-Supervisor:

.....KR. 2025/08/30.....

Kaushalya Rajapaksha

Assistant Lecture at SLIIT

Figure : Dataset Requesting letter 1 (SLTB)

The Managing Director

Ashok Leyland Lanka (Pvt) Ltd

Request for Dataset Access for Research Project on "Safety Monitoring System for Public Transport in Sri Lanka"

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Name: .....

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Thisara Shyamalee

Lecture at SLIIT

Co-Supervisor:

.....KR.....2025/08/30.....

Kaushalya Rajapaksha

Assistant Lecture at SLIIT

Figure : Dataset Requesting letter 2 (Lanka Ashok Leyland)